

Lunar Greenhouse

Campaign 1 · Case 02 · Shackleton Crater, Lunar South Pole



SOLAR
AGRICULTURAL
AGENCY
Accessible
Mission

MISSION QUESTION

Why do healthy lunar tomato plants form flowers but no fruit?

YOUR ROLE

Process Modeler. Work one step at a time. You may answer in phrases or bullet points.

 TASK 01 **1 · Vocabulary**

Anther	Structure that produces and holds pollen.
Fertilization	Joining of reproductive cells at an ovule.
Fruit set	Beginning of fruit development.
Pollen tube	Growth from a pollen grain through the style.
Pollination	Transfer of pollen to a receptive stigma.
Stigma	Receptive surface where pollen must arrive.

 TASK 02 **2 · Initial thinking**

A plant is healthy and flowering but produces no fruit. Which reproductive step would you investigate first? What evidence would you seek?

SCIENCE FOCUS

Pollination, fertilization, and fruit set are different events. A failure early in the sequence can prevent every later event.



TASK 03 3 · Model the pollination sequence

Use each phrase once. Write one phrase in each numbered box.

WORD BANK fruit set · physical agitation · viable pollen in anthers · fertilization · pollen tube growth · pollen reaches stigma

Step 1

Step 2

Step 3

Step 4

Step 5

Step 6

TEXT MODEL RULE

Step 1 enables Step 2. Step 2 enables Step 3. Continue in order. When the first step fails, later steps are interrupted.



TASK 04 4 · Identify the failed step

Use the case evidence to locate the first failure. Answer in phrases or bullet points.

Failed step

Observation 1

Observation 2

One downstream result



TASK 05 5 · Test the competing explanations

For every diagnosis, write Supported or Weakened. Then explain one evidence-based reason.

Lunar regolith is toxic to fruit development.

Classification:

Evidence-based reason

The light spectrum lacks ultraviolet light needed for fruiting.

Classification:

Evidence-based reason

No effective pollen release and transfer is occurring.

Classification:

Evidence-based reason

Carbon dioxide is too high for tomatoes.

Classification:

Evidence-based reason



TASK 06

6 · Diagnose and reject an alternative

Diagnosis: state the cause and the failed biological step.

Rejected alternative: name it and explain which evidence weakens it.



TASK 07

7 · Claim-Evidence-Reasoning

Claim

Evidence: two or three specific pieces

Reasoning: connect the evidence to the process model



TASK 08 8 · Design a reliable pollination support

Design

One criterion

One constraint

Mechanism check: restored step and evidence of success



TASK 09 9 · Exit ticket

A second sealed habitat grows a different self-pollinating crop. Flowers and viable pollen are present, but no fruit forms. Which step should investigators test first, and what observation would show whether it is failing?